

GRATES COVE, NL, Canada

WMO: 713360

Lat: 48.1700N

Lon: 52.9400W

Elev: 152

StdP: 14.62

Time Zone: -3.50 (NAF)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.1	12.1	0.7	5.7	11.3	3.8	6.7	13.8	51.3	25.3	47.6	29.5	20.6	300	0.944

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.4	73.2	64.9	70.8	64.1	68.6	63.1	67.6	70.7	66.0	69.1	64.5	67.4	18.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.4	97.9	69.5	64.8	92.2	68.0	63.1	87.0	66.5	32.2	71.0	30.9	69.3	29.7	67.5	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
42.2	36.9	33.8	4.1	78.7	5.2	2.5	0.3	80.5	-2.8	82.0	-5.7	83.4	-9.5	85.2
			4.6	70.9	3.8	2.1	1.9	72.4	-0.4	73.6	-2.5	74.8	-5.2	76.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.0	26.7	25.8	28.2	35.0	41.7	50.3	60.0	61.7	55.6	47.1	39.6	31.3
	DBStd	13.84	7.15	7.20	6.25	5.23	6.08	6.72	6.02	4.50	5.39	5.60	6.07	6.78
	HDD50	3916	724	679	675	449	269	79	5	0	11	127	317	581
	HDD65	8430	1189	1099	1140	899	722	442	172	119	285	555	762	1046
	CDD50	998	0	0	0	0	12	88	315	362	179	37	5	0
	CDD65	36	0	0	0	0	0	0	17	16	3	0	0	0
	CDH74	43	0	0	0	0	0	0	24	17	2	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	17.1	20.4	19.3	18.5	16.6	14.4	14.1	15.0	14.9	16.5	17.6	18.7	19.9
Precipitation	PrecAvg	55.9	5.5	5.0	4.6	4.2	3.8	3.8	4.0	3.9	4.8	5.5	5.3	5.6
	PrecMax	66.7	9.2	7.2	8.3	9.4	6.0	5.7	6.4	6.3	8.8	9.5	8.3	9.5
	PrecMin	44.3	3.1	2.6	2.0	1.8	1.6	1.9	2.5	1.9	2.1	1.7	2.4	2.4
	PrecStd	5.4	1.6	1.1	1.3	1.7	1.1	1.0	1.0	1.1	1.6	1.8	1.4	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.1	47.7	49.4	54.9	64.0	71.8	76.9	76.2	72.3	65.9	59.8	54.6
		MCWB	49.2	46.9	46.6	48.8	54.4	63.0	65.7	66.7	65.8	63.2	57.8	53.2
	2%	DB	44.6	41.8	43.5	49.7	59.3	67.9	73.7	72.9	68.3	61.9	55.7	49.5
		MCWB	43.7	41.1	41.4	46.0	52.0	60.0	64.8	65.1	63.0	59.6	54.3	48.4
	5%	DB	39.3	38.1	38.8	45.9	55.4	64.4	71.1	70.6	65.9	58.6	52.3	44.2
		MCWB	38.2	37.4	37.1	42.8	49.8	58.1	64.1	64.5	61.9	56.4	51.0	43.0
	10%	DB	35.8	34.7	35.7	42.4	51.7	61.1	68.8	68.6	63.7	55.5	48.8	39.9
		MCWB	34.8	33.7	34.0	39.2	46.7	56.2	63.1	63.7	60.0	52.9	47.0	38.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.5	48.5	47.3	51.9	58.3	65.7	69.9	70.0	67.3	64.2	58.2	53.9
		MCDB	50.3	48.9	48.3	53.4	62.1	70.0	74.0	72.7	70.0	65.9	59.4	54.8
	2%	WB	44.0	42.0	41.7	47.3	54.4	62.5	67.6	67.9	64.9	60.6	54.6	48.3
		MCDB	44.7	42.5	43.0	49.0	57.8	66.1	71.3	70.5	67.4	61.8	55.7	49.4
	5%	WB	38.4	38.1	37.1	43.5	50.6	59.8	66.1	66.1	63.1	56.9	51.3	43.2
		MCDB	39.0	38.7	38.1	45.6	55.3	62.7	69.5	68.8	65.5	58.5	52.5	44.0
	10%	WB	34.7	34.4	34.3	40.2	47.4	56.8	64.4	64.5	60.9	53.6	47.7	38.6
		MCDB	35.5	35.0	35.4	42.2	51.8	60.6	67.8	67.3	63.5	55.4	48.9	39.7
Mean Daily Temperature Range	5% DB	MDBR	10.4	9.3	9.0	9.2	11.3	12.5	12.4	10.4	9.9	7.9	8.4	8.7
		MCDBR	17.5	14.7	14.6	15.5	18.6	18.5	15.5	13.7	12.8	12.2	14.0	16.0
		MCWBR	17.2	14.7	13.3	12.3	13.1	11.7	9.3	8.4	9.3	11.3	14.1	16.9
	5% WB	MCDBR	16.9	14.4	13.6	14.9	17.7	16.5	13.7	12.3	12.7	12.6	14.1	15.6
		MCWBR	17.5	15.0	13.4	13.2	13.6	12.9	9.7	9.2	10.5	12.3	14.3	16.6
Clear-Sky Solar Irradiance	taub	0.317	0.313	0.337	0.367	0.372	0.403	0.428	0.399	0.373	0.348	0.323	0.311	
	taud	2.325	2.417	2.356	2.350	2.416	2.354	2.331	2.418	2.463	2.476	2.454	2.354	
	Ebn at Noon	234	269	279	280	282	273	264	267	264	252	234	221	
	Edh at Noon	23	26	33	36	35	38	38	34	30	25	20	20	
All-Sky Solar Radiation	RadAvg	298	548	902	1266	1521	1628	1650	1388	1084	646	364	237	
	RadStd	24	44	84	125	121	129	140	91	72	66	48	22	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page